

Are large language models like ChatGPT really that harmful to the environment?

Posted by u/ropika4 • 0 points • 35 comments

I've heard that training these kinds of AI systems consumes a huge amount of energy, water, and leads to significant carbon emissions.

Some sources even claim that they are quite harmful to the environment.

Are there any scientifically accepted studies or data on this? Can we really say they have such a major environmental impact?

Comments

u/FunctionFluffy4932 • 1 points • 2026-02-06 10:22 UTC

I saw an article recently that said watching one hour on Netflix creates 500 times more CO² than two text prompts in ChatGPT or Gemini do. So Idk what to believe about how bad AI is for the environment?

u/[deleted] • 1 points • 2025-11-23 01:24 UTC

[removed]

u/the_anonymizer • 1 points • 2025-09-11 15:40 UTC

brain is not harmful to the environment. Solution is : using our brains. However, if AI becomes less harmful in the near term than using classical computer science + people's brain engineering etc, then ok, AI is an investment. I think AI can save the World, if we use it to save the World. If we use it only to make money, the problem might be huge for the planet. It all depends on our own goal as mankind. May mankind be guided in the path of wisdom with this technology, because with great power, great things can be done, but also very bad things. I hope mankind can be responsible enough to save itself from its own greed and its impact destroying our own planet.

u/iboughtarock • 1 points • 2025-06-06 17:11 UTC

It is basically a non-issue once you look at the numbers.

****Domestic/municipal uses**** (drinking, showers, cooking, toilet flushes, laundry, etc.): 133 trillion gal yr⁻¹

****Data-centre cooling**** (on-site evaporation + blow-down): 43.7 billion gal yr⁻¹

Both look like big numbers, but if you drop them to the correct scale $43.7 / 133,000 = 0.0328\%$ of global water usage is from data centers.

u/JudgeMyReinhold • 1 points • 2026-01-18 03:34 UTC

You really gonna lump drinking water into this? There's a difference between using water to literally survive vs using AI and cooling data centers.

u/iboughtarock • 1 points • 2026-01-18 23:09 UTC

So using AI to solve novel problems in science is considered useless? The point is *scale*. Its less than 1/10th of a percent.

u/JudgeMyReinhold • 1 points • 2026-01-19 00:35 UTC

You're right, scale is important. So maybe you should focus on data center water usage relative to the towns those data centers are nearest.

u/iboughtarock • 1 points • 2026-01-19 15:51 UTC

I guess just the way you phrased it and the context of the thread, I thought it was implied.

Yes I agree with you there. Drying up aquifers and evaporating them into the atmosphere is [a major problem](<https://thehill.com/policy/equilibrium-sustainability/4426143-majority-of-americas-underground-water-stores-are-drying-up-study-finds/>).